AI IN CYBERSECURITY

Using Artificial Intelligence for Real-Time Threat Detection, Anomaly Identification and Automated Response in a Financial Services Company

Submitted To : Mr. Chintan Patel

Submitted by: Naisargi Thakkar

Roll No. / Enrolment No.: AIDS20262006

Subject: Introduction to AI

1. Understanding the Problem

A financial services company, such as a bank, an insurance company or a payments firm, looks after two things that criminals badly want: people's money and people's personal data. As more services move online (mobile banking, UPI payments, digital loans), the number of "doors" that attackers can try also grows.

The problem in the brief is simple to state. The company is facing more cyber threats and smarter attacks. The question is: **how can Artificial Intelligence (AI) help with (a) real-time threat detection, (b) spotting unusual activity (anomalies) and (c) automatic response, so that sensitive customer data stays protected?** The brief is reproduced in Figure A1 in the Appendix.

The cost of failure is high. IBM's 2025 study found that the average data breach in India cost ₹220 million, up 13% from the year before, and that 73% of the Indian organisations studied used AI and security automation only a little or not at all [1]. Globally, a breach in the financial industry cost about US$5.56 million on average, against US$4.44 million for all industries [2].

Why are older methods not enough? Traditional security is like a guard who carries photos of known thieves. If a thief changes his face, the guard misses him. AI is more like an experienced guard who notices strange behaviour, such as someone walking around the vault at 3 a.m., even if he has never seen that person before. This report explains how that works, what can go wrong, what Indian rules say, and how three companies (Palo Alto Networks, Quick Heal and Microsoft) use AI. All images, charts and tables are placed in the Appendix.

2. Major Cyber Security Threats Faced by a Financial Services Company

A financial company faces many kinds of attacks at the same time. The most important ones are:

- **Phishing and social engineering:** fake emails, SMS or calls that trick staff or customers into sharing passwords or OTPs. This is a very common starting point for bigger attacks.
- **Ransomware:** malware that locks a company's files and demands money. New versions are built to avoid old "signature-based" detection [3].
- **Account takeover:** criminals log in as real customers using stolen or guessed passwords.
- **Malware and fake apps:** programs that quietly steal login details from phones and computers.
- **Insider threats:** an employee or contractor who misuses access, on purpose or by mistake.
- **Third-party (supply-chain) risk:** a weakly protected vendor becomes a back door into the company.
- **DDoS attacks:** a flood of fake traffic that crashes websites and apps so customers cannot use them.
- **AI-powered fraud:** deepfake voices and perfectly written scam messages (see Section 8).

The scale is huge. CERT-In reported over 1.59 million cyberattacks in India in 2023 [4]. No human team can read every alert from so many sources, which is exactly where AI helps.

3. Data Privacy Concerns in AI Implementation and Protecting Sensitive Customer Data

A financial company holds names, addresses, ID details, account numbers and every transaction a customer has made. AI needs a lot of data to learn, and that creates new privacy risks:

- **Collecting too much data** "just in case" makes a bigger target for thieves.
- **Data leaving the company's control,** for example when it is sent to an outside cloud or AI vendor, or when staff paste customer details into a public chatbot ("shadow AI").
- **Models that remember too much,** which can accidentally reveal private details from the training data.
- **Using data for a purpose the customer never agreed to.**

IBM found that in India only 37% of organisations had access controls for AI, and that shadow AI added about ₹17.9 million to the average cost of a breach [1]. NIST also treats privacy as a core part of trustworthy AI [5].

Simple protection steps include: collect only what is needed; hide or mask personal details before training a model; encrypt data when stored and when sent; allow only the right people to see it and keep a record of who did; check vendors carefully; and, where possible, use methods such as federated learning, which lets a model learn without moving raw customer data to one place. India's DPDP Act 2023 adds legal duties on consent and safeguards (Section 12).

4. Real-Time Threat Detection Using AI/ML: Techniques, Workflow and Use Cases

"Real time" means the system judges each event within seconds, not the next morning. The usual workflow is shown in Figure A2 (Appendix):

1. **Collect data** from logins, transactions, emails, devices and networks.
2. **Clean and prepare** the data (remove errors, mask private details).
3. **Run AI/ML models** that look for known and unknown threats.
4. **Give a risk score** (low, medium or high) to every event.
5. **Act automatically** on high-risk events, and send unclear ones to a human.
6. **Learn:** analysts' decisions are fed back so the models improve.

Main techniques. Supervised learning learns from past examples labelled "fraud" or "not fraud". Unsupervised learning finds odd behaviour without labels, which helps against brand-new attacks. Deep learning handles complex patterns such as new malware or phishing text. Behaviour-based detection watches what a program does instead of what it looks like. Quick Heal's patented anti-ransomware tool works this way, blocking new ransomware without needing a signature [3].

Use cases: catching phishing emails, blocking ransomware, stopping card and UPI fraud, flagging suspicious logins and spotting intruders on the network. Modern platforms collect data from firewalls, cloud services, laptops and identity systems in one place [6], and use ML-enhanced rules to find threats and reduce false alarms [7].

5. Anomaly Identification Using ML: Behavioural Analytics and Unusual Transaction Patterns

An anomaly is something that does not fit normal behaviour. The AI first learns what "normal" looks like for each customer and employee: usual login time, device, city, typical amount and usual payees. When something deviates, it is flagged [8]. This is called user and entity behaviour analytics (UEBA), and Palo Alto's XSIAM includes it [6]. Figure A5 (Appendix) shows a sample fraud-monitoring dashboard.

Example: A customer normally sends ₹2,000–₹5,000 in the evening from one phone in Vadodara. At 3 a.m., a new device in another state tries to send ₹2 lakh to a new payee, then tries again. Each step alone may look harmless, but together they create a high risk score. The bank can then ask for extra verification or hold the payment.

Common ML methods are statistical limits (for example "more than 5 times the usual amount"), clustering (grouping similar customers and checking who falls outside), Isolation Forest (finding points that are easy to "isolate") and autoencoders (neural networks that struggle to rebuild unusual data) [9]. An important point: an anomaly is not always an attack. A customer may simply be on holiday. That is why good systems combine many signals and use a risk score instead of a simple yes/no.

6. AI-Powered Automated Incident Response and Threat Containment

Detecting a threat is only half the job. Containment means stopping it from spreading. Attackers can move in minutes, so waiting for a person to read an alert may be too slow. Automated response (often called SOAR) follows ready-made "playbooks", such as:

- isolating an infected laptop from the network;
- disabling a hacked account or forcing a password reset;
- blocking a bad IP address or website;
- holding a suspicious payment until the customer confirms; and
- opening a ticket and alerting the security team, with the evidence attached.

Microsoft gives a financial example: when its system detects fraud techniques in a connected SAP system, it can automatically disable the user's accounts to stop money being transferred, without waiting for the security team [10]. Palo Alto says XSIAM handles data collection, analysis and first checking of alerts, so analysts only look at incidents that need a human [11]. IBM reports that organisations using AI and automation heavily saved US$1.9 million per breach and shortened the breach lifecycle by 68 days [2]. Automation can also help meet India's rule to report incidents to CERT-In within six hours [4].

Safety rule: low-risk actions (blocking one suspicious email) can be fully automatic, but high-impact actions (shutting down a payment system) should need human approval.

7. Case Studies: AI-Driven Financial Cybersecurity Solutions

7.1 Palo Alto Networks (global)

Palo Alto Networks' main AI product for security teams is **Cortex XSIAM**, launched in 2022. It joins XDR (detection across devices and networks), the automation of XSOAR, AI/ML, user behaviour analytics and attack-surface management in one cloud platform [6]. A Fortune 500 bank case study describes how the bank combined many SOC tools into one platform, used automation to reduce manual work, and met strict compliance needs; the platform is SOC 2 Type II certified [12]. In a vendor white paper on six customers (not all in finance), Palo Alto reports incident resolution falling from days to minutes [13]. These results are reported by the vendor, so they should be read with some care. A screenshot of the XSIAM command centre is in Figure A6.

7.2 Quick Heal / Seqrite (India)

Quick Heal Technologies, based in Pune, sells to consumers under the Quick Heal brand and to businesses under **Seqrite** [14]. Its AI engine, **GoDeep.AI**, mixes deep learning, behaviour detection, predictive detection and AI [15]. Quick Heal was the first Indian company to receive a US patent for AI/ML-based, signature-less anti-ransomware technology that blocks new ransomware in real time [3]. It also offers AntiFraud.AI for fraud protection [15] and highlights its large malware-analysis lab for local threat knowledge [16]. In our view, its

strength is endpoint protection and India-specific threat research, which suits small and mid-sized firms; large banks usually also need a wider SOC platform. A screenshot of Seqrite's cloud console is in Figure A7.

7.3 Microsoft (global)

Microsoft combines **Microsoft Sentinel** (a cloud SIEM that collects security data), **Defender XDR** and **Security Copilot** (a generative-AI assistant). Microsoft says its AI-based detection, correlation and investigation reduce false positives and the time to respond [7]. Security Copilot is sold on a usage-based model measured in Security Compute Units [17], and its "attack disruption" can automatically stop financial fraud in connected SAP systems [10]. It fits best where a company already uses Microsoft tools. A screenshot of Defender with Security Copilot is in Figure A8, and a side-by-side comparison is in Table A9 (Appendix).

8. Adversarial AI, AI-Enabled Cyberattacks and Defensive Countermeasures

AI helps defenders, but attackers use it too. There are two separate problems:

- **Attackers using AI:** writing convincing phishing messages in any language, cloning a manager's voice or face (deepfakes) to order a payment, scanning for weak points automatically, and building malware that keeps changing.
- **Attacks on the bank's own AI:** evasion (slightly changing an input so the model is fooled), data poisoning (slipping bad data into training so the model learns wrong lessons), model theft (copying a model by asking it many questions) and prompt injection (hiding instructions to trick chatbots and AI assistants) [18].

Defences include: training models with tricky examples (adversarial training); checking and cleaning input data; protecting the training data and limiting who can use or change models; watching for sudden drops in model accuracy; testing the system with friendly "red team" attackers; and keeping humans in the loop for risky actions, such as a call-back to a known number before any large payment. Strong multi-factor authentication also reduces the damage when a voice or password is faked. Since many organisations lack AI controls [1], securing the AI itself is now part of cybersecurity.

9. Model Performance:

An AI model performance will depends on how you will feed the data. If the data is not latest , incomplete, or full of errors, the model will make wrong decisions. Fraud is also rare, so

model can look 99% accurate by simply calling everything normal. Therefore accuracy alone can measure the performance of ypur model.

Two critical things are. A false positive is a false alarm: a genuine customer is blocked or an analyst wastes time. A false negative is a missed attack, which is usually more dangerous. Reducing one often increases the other, so the bank must choose a balance.

Figure A4 (Appendix) gives a simple example. The key metrics are:

•Precision = correct alerts ÷ all alerts = 80 ÷ 280 = about 29%. (Of every 100 alerts, about 29 are real.)

•Recall = frauds caught ÷ all frauds = 80 ÷ 100 = 80%.

•F1 score = the balance of precision and recall (about 0.42 here).

•Operational metrics: mean time to detect and respond, and analyst workload.

Dashboards such as Figure A5 show these metrics live. Models also "drift" as criminals change tactics, so models must be tested regularly, compared with simple threshold rules, and should be retrained with fresh data as per requirements.

10. Explainable AI, Model Transparency and Auditability in Financial Cybersecurity

Some AI models work like a "black box": they give an answer but not the reason. In finance this is risky. A customer will ask, "Why was my payment blocked?", and auditors and regulators will ask, "How did the system decide?". Explainable AI (XAI) gives reasons in plain terms, such as "new device, unusual amount and new payee".

Useful tools are SHAP and LIME, which show how much each factor pushed a decision [19, 20]; simpler models where possible; model documents that record purpose, data and limits; and full logs of inputs, model version and outcome for every decision, so the case can be checked later (auditability). The RBI's FREE-AI report likewise stresses board-approved AI policies, audits and incident reporting [21, 22].

11. Ethical AI, Bias, Accountability :

When you feed unbalanced data to your model , there is an defiantly chance of bias. For example, a fraud model might treat customers from certain regions, or those with little digital history, as "risky" and block them more often. Such unfair blocking hurts customers and the company's reputation. To reduce bias: check results across different customer groups, use varied data, and review complaints.

Accountability means a named person or team is responsible for each AI system. A bank cannot blame the vendor's software when something goes wrong. Human oversight means people can review, override and switch off the AI, and customers have an easy way to appeal. The RBI's FREE-AI framework is built on seven guiding principles ("sutras"), one of which says trust is the foundation for a sector that safeguards people's money [21, 22].**Accountability** means a named person or team is responsible for each AI system. A bank cannot blame the vendor's software when something goes wrong. **Human oversight** means people can review, override and switch off the AI, and customers have an easy way to appeal. The RBI's FREE-AI framework is built on seven guiding principles ("sutras"), one of which says trust is the foundation for a sector that safeguards people's money [21, 22].

12. Indian Financial Cybersecurity Regulations and Compliance Requirements

Several rules apply together (summarised in Table A10, Appendix):

- **RBI Cyber Security Framework for banks (2016):** banks must report cyber incidents to the RBI within 2–6 hours of detection [23].
- **RBI Master Directions (2023):** one on IT governance, risk, controls and assurance (effective 1 April 2024) and one on outsourcing of IT services, which matters when using outside AI or security vendors [24].
- **DPDP Act 2023:** requires consent, reasonable security safeguards and breach reporting; fines can reach ₹250 crore, and full enforcement is expected from 13 May 2027 [25]. Its rules ask for a full breach report within 72 hours [26].
- **SEBI CSCRF (August 2024):** a cyber-resilience framework for SEBI-regulated entities [24].
- **RBI FREE-AI report (August 2025):** a committee report with 7 principles and 26 recommendations in six pillars, including board-approved AI policies [21, 27]. It is guidance, so companies should watch for formal rules.

One incident can trigger several reports at once (CERT-In, RBI and the Data Protection Board), so a single, well-practised response plan is needed. Rules are changing, so the latest official notifications should always be checked.

13. Overcoming AI Implementation Challenges: Responsible AI Integration

Common challenges are poor or scattered data, old systems that do not connect, a shortage of skilled people, too many false alarms, dependence on one vendor, high cost and staff who do not trust the AI. IBM also found that nearly 60% of Indian organisations lacked fully developed AI governance policies [28]. A practical path is:

1.Start with small pilot project , small implementation for one use case like fraudulent alert.

2.Check whether any missing data values or wrong datatypes cleaning process and extract useful features from the available data

3.Consider AI governance: a policy, an owner, and approval for new models.

4.Roll out in stages: AI only suggests, then a human approves, then low-risk actions become automatic.

5.Check vendors for security, data location and exit options, as required under RBI outsourcing rules.

6.Training of employees and keep monitoring accuracy, bias and drift.

The NIST AI Risk Management Framework offers a simple structure for this: Govern, Map, Measure and Manage [5].

14. Cost-Benefit Analysis and ROI of AI Implementation

Costs: software licences or usage fees (for example, Security Copilot is billed by usage [17]), data storage and collection, linking old systems, skilled staff, training, regular tuning, and compliance work. **Benefits:** cheaper and shorter breaches (IBM's US$1.9 million average saving and 68 fewer days [2]), less analyst time wasted on routine alerts, fewer fraud losses, avoiding penalties (up to ₹250 crore under DPDP [25]) and stronger customer trust. The full working is in Tables A12 to A15 (Appendix).

15. Conclusion and Recommendations

AI gives financial companies a faster and smarter way to defend against modern attacks. It detects threats in real time, spots unusual behaviour and acts within seconds, as the Palo Alto, Quick Heal and Microsoft examples show. But AI is not magic. It needs clean data, privacy protection, regular testing, explainable decisions, human oversight and compliance with Indian law. The company should: (1) start with a focused pilot; (2) protect customer data from day one; (3) keep humans in charge of high-impact decisions; (4) measure false positives and false negatives, not just accuracy; and (5) follow RBI, CERT-In and DPDP requirements in one joined-up plan.

References

[1] IBM India Newsroom (7 Aug 2025). India Records Highest Average Cost of a Data Breach at INR 220 million in 2025: IBM Report. https://in.newsroom.ibm.com/2025-08-07-India-Records-Highest-Average-Cost-of-a-Data-Breach-IBM

[2] IBM (2025). Cost of a Data Breach Report 2025: key findings (as published by Northdoor). https://www.northdoor.co.uk/?p=35370

[3] CRN India. Quick Heal Technologies granted US patent for anti-ransomware technology. https://www.crn.in/?p=6946

[4] ManageEngine. CERT-In compliance: How to meet the 6-hour incident reporting deadline and avoid penalties. https://www.manageengine.com/compliance-manager/cert-compliance/

[5] NIST (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0), NIST AI 100-1. https://www.nist.gov/itl/ai-risk-management-framework

[6] World Wide Technology. How Palo Alto Networks is revolutionizing security operations with XSIAM. https://www.wwt.com/article/how-palo-alto-networks-is-revolutionizing-security-operations-with-xsiam

[7] Microsoft. Microsoft Sentinel: AI-powered SIEM and XDR (product page). https://www.microsoft.com/es-mx/security/business/siem-and-xdr/microsoft-sentinel

[8] Star Knowledge. How Microsoft Sentinel Uses AI to Detect and Respond to Threats Faster. https://star-knowledge.com/blog/how-microsoft-sentinel-uses-ai-to-detect-and-respond-to-threats-faster/

[9] Chandola, V., Banerjee, A., & Kumar, V. (2009). Anomaly detection: A survey. ACM Computing Surveys, 41(3).

[10] Mastering SAP (5 Dec 2023). Microsoft's New Security Copilot Shields SAP Systems with AI. https://masteringsap.com/blogs/microsofts-new-security-copilot-shields-sap-systems-with-ai/

[11] Palo Alto Networks. Form 10-K, fiscal year 2026 (U.S. SEC filing). https://www.sec.gov/Archives/edgar/data/0001327567/000132756726000023/panw-20260731.htm

[12] Palo Alto Networks. Cortex XSIAM reshapes SecOps for Fortune 500 financial giant (customer story). https://www.paloaltonetworks.com/customers/cortex-xsiam-reshapes-secops-for-fortune-500-financial-giant

[13] Palo Alto Networks (Oct 2024). AI-Driven SOC Transformation with Cortex XSIAM (white paper summary). https://bitpipe.computerweekly.com/detail/RES/1741763121_685.html

[14] Indian Link. Quick Heal 1st Indian firm to get US patent for anti-ransomware tech. https://indianlink.com.au/?p=88009

[15] Quick Heal Technologies Ltd. (July 2025). Quick Heal rolls out Internet Security Essentials in Tamil Nadu (press release). https://quickheal.co.in/documents/media/2025/qh-rolls-out-internet-security-essentials-in-tamil-nadu-digital-terminal-july2025.pdf

[16] Quick Heal. 7 Reasons why you need Quick Heal to combat Financial Frauds. https://www.quickheal.com/blogs/7-reasons-why-you-need-quick-heal-to-combat-financial-frauds/

[17] Inventive HQ. Microsoft Security Copilot (pricing and features overview). https://inventivehq.com/software/security-copilot

[18] NIST (2025). Adversarial Machine Learning: A Taxonomy and Terminology of Attacks and Mitigations, NIST AI 100-2 E2025. https://csrc.nist.gov/pubs/ai/100/2/e2025/final

[19] Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems (NeurIPS).

[20] Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?" Explaining the predictions of any classifier. Proceedings of KDD 2016.

[21] Reserve Bank of India (Aug 2025). FREE-AI Committee Report: Framework for Responsible and Ethical Enablement of Artificial Intelligence. https://rbi.org.in/SCRIPTS/PublicationReportDetails.aspx?ID=1306

[22] Mondaq. RBI Releases Report On Responsible And Ethical Enablement Of AI In The Financial Sector. https://admin.mondaq.com/india/new-technology/1670830/rbi-releases-report-on-responsible-and-ethical-enablement-of-ai-in-the-financial-sector

[23] Baker McKenzie. Global Data and Cyber Handbook, India: Security Requirements and Breach Notification. https://resourcehub.bakermckenzie.com/en/resources/global-data-and-cyber-handbook/asia-pacific/india/topics/security-requirements-and-breach-notification

[24] CyberSigmacs. Compliance registry (RBI, SEBI, DPDP summaries). https://cybersigmacs.com/compliance-registry/

[25] CyberSigmacs. India cybersecurity compliance deadlines 2026. https://cybersigmacs.com/research/india-cybersecurity-compliance-deadlines-2026/

[26] Matters.ai. DPDP breach notification. https://www.matters.ai/article/dpdp-breach-notification

[27] Lakshmikumaran & Sridharan. RBI framework: Responsible and ethical enablement of AI in financial sector. https://cn.lakshmisri.com/insights/articles/rbi-framework-responsible-and-ethical-enablement-of-artificial-intelligence-in-financial-sector

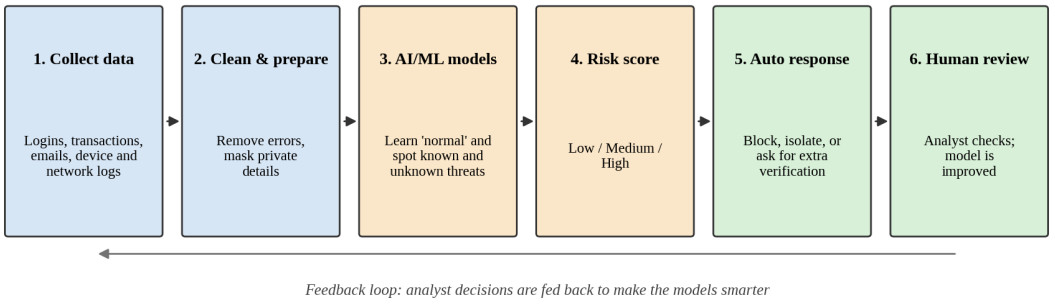
[28] CRN India (2025). India records highest average cost of data breach at ₹220 Mn in 2025: IBM Report. https://www.crn.in/?p=72635

APPENDIX

Figure A1: Problem statement given in the assignment brief

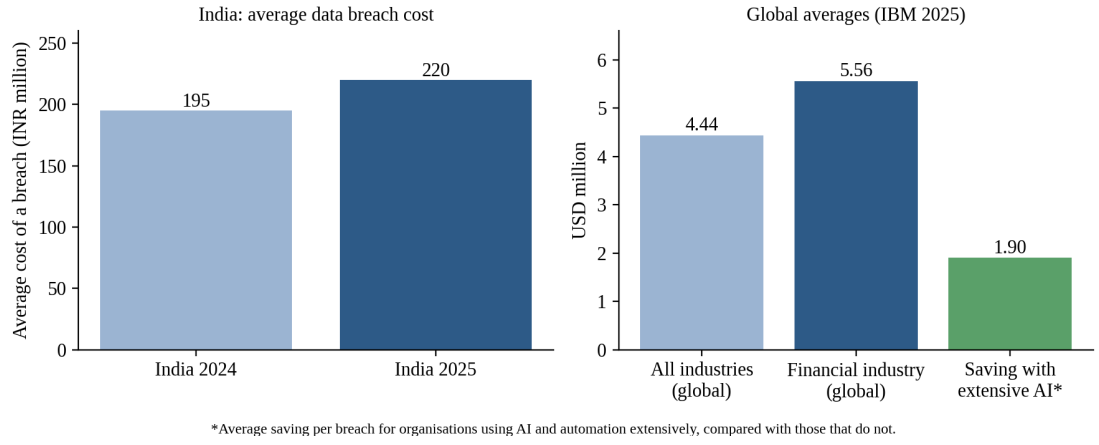
- AI in Cybersecurity: A financial services company is facing increasing cyber threats and sophisticated attacks. How can AI be used for real-time threat detection, anomaly identification, and automated response to protect sensitive customer data?
- Quick Heal (India)
- Palo Alto Networks (Global)
- Microsoft (Global)

Figure A2: How AI detects and responds to a cyber threat (workflow)



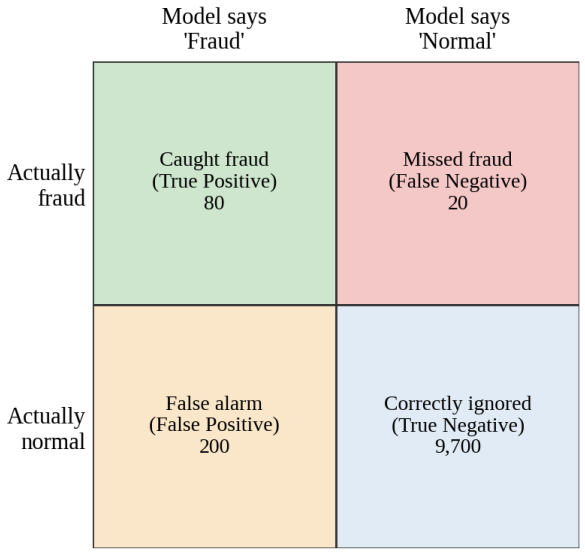
Source: AI-generated diagram (created by Claude, an AI assistant by Anthropic, using Python).

Figure A3: Cost of data breaches and savings from AI (IBM 2025)



Source: Chart generated by Claude (AI) using Python. Data from IBM Cost of a Data Breach Report 2025 [1, 2].

Figure A4: Example of false positives and false negatives (illustrative)



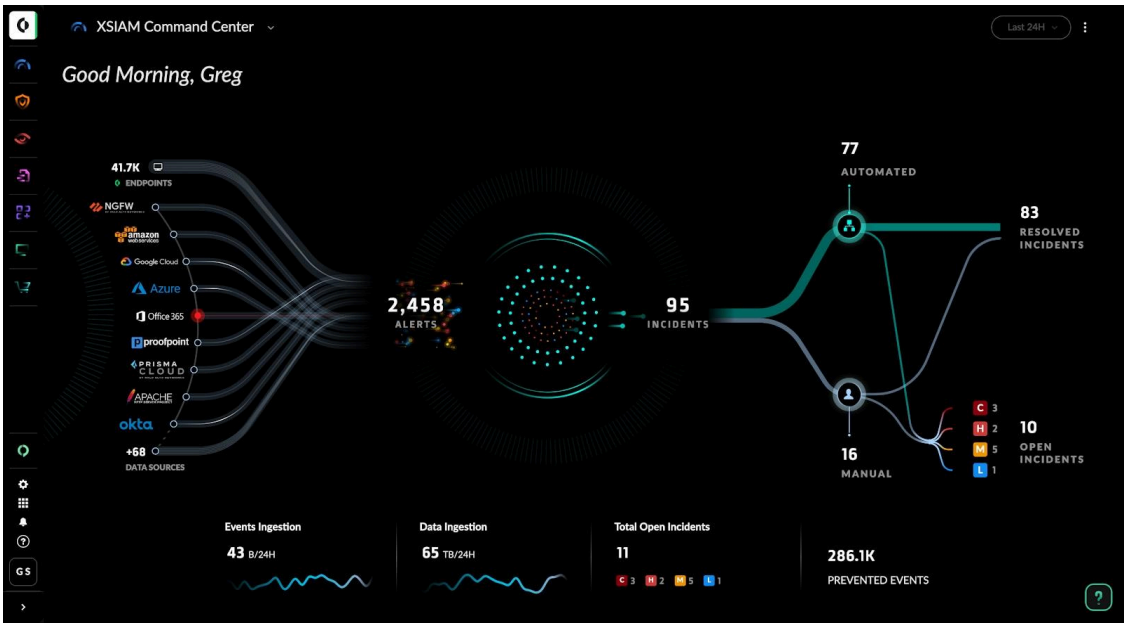
Source: AI-generated chart (created by Claude using Python). The numbers are made up for explanation only and are not real bank data.

Figure A5: Example of a real-time fraud detection dashboard (sample data)



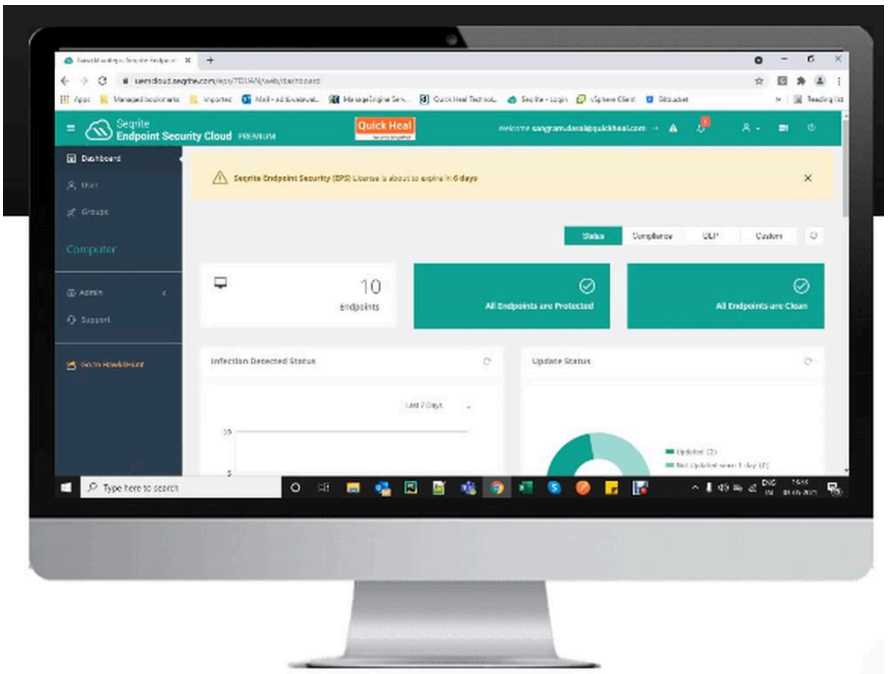
Source: https://byteboon.com

Figure A6: Palo Alto Networks Cortex XSIAM "Command Center"



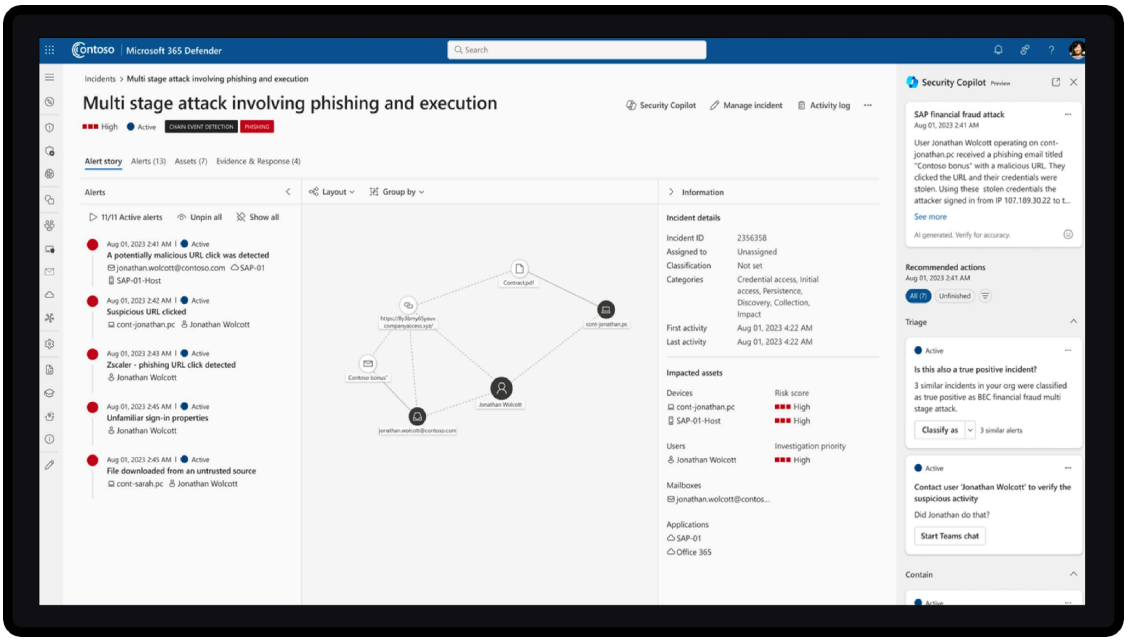
Source: [Leader in Cybersecurity Protection & Software for the Modern Enterprises - Palo Alto Networks](#) The product is described in [11].

Figure A7: Quick Heal Seqrite Endpoint Security Cloud dashboard



Source: [Compare B2B Software, Download, & Develop Open Source & Business Software - SourceForge](#)

Figure A8: Microsoft 365 Defender with Security Copilot (SAP financial fraud example)



Source: https://www.microsoft.com "Contoso" is Microsoft's demo company, so the data is sample data. The product is described in [7, 10].

Table A9: Comparison of the three case studies

Point	Palo Alto Networks	Quick Heal (Seqrite)	Microsoft
Origin	USA; global	Pune, India; founded 1995 [14]	USA; global
Main AI products	Cortex XSIAM, XSOAR, XDR [11]	Quick Heal and Seqrite; GoDeep.AI; AntiFraud.AI [15]	Sentinel, Defender XDR, Security Copilot [7]
How AI is used	Collects data from many sources, joins alerts into incidents, automates checking and response [6]	Deep learning and behaviour-based detection; blocks unknown ransomware [3]	ML-enhanced detection, behaviour analytics, AI-guided investigation; attack disruption [7, 10]
Strengths	All-in-one SOC platform; strong automation	India-focused threat research; patented anti-ransomware tech	Large ecosystem; cloud-based; works with other Microsoft tools
Points to watch	Many results are vendor-reported; can be complex and costly	In our view, stronger on endpoints and smaller firms than on large-bank SOCs	Best value if already on Microsoft; usage-based pricing [17]

Table A10: Indian rules that apply to AI-driven cybersecurity in finance

Rule	Applies to	Key point
RBI Cyber Security Framework in Banks (2016)	Banks	Board-approved cyber policy, SOC; report incidents to RBI within 2–6 hours [23]

Rule	Applies to	Key point
RBI Master Direction on IT Governance, Risk, Controls and Assurance (2023)	Banks, NBFCs and others	IT governance and controls; effective 1 April 2024 [24]
RBI Master Direction on Outsourcing of IT Services (2023)	Regulated entities	Care when using outside AI or security vendors; effective 1 October 2023 [24]
CERT-In Directions (April 2022)	All organisations	Report cyber incidents within 6 hours [23]
DPDP Act 2023 and Rules	All data fiduciaries	Consent, safeguards, breach notice; penalties up to ₹250 crore; full enforcement 13 May 2027 [25]
SEBI CSCRF (20 Aug 2024)	SEBI-regulated entities	Cyber-resilience framework [24]
RBI FREE-AI Report (Aug 2025)	Financial sector (recommendations)	7 principles, 26 recommendations, six pillars; board-approved AI policy [21, 27]

Table A11: Yearly costs of the AI security platform (illustrative, ₹ crore)

Cost item	How the figure is worked out (assumed)	Cost (₹ crore)
Software licences / usage fees	Platform subscription and usage-based charges	2.00
Data storage and log collection	Cloud storage for security logs and transaction data	0.80
Linking old systems	Integration work with existing banking systems	0.70
Skilled staff	3 extra security / data specialists × ₹30 lakh each = ₹90 lakh	0.90
Training	Staff and analyst training	0.20
Tuning and monitoring	Testing the model, fixing false alarms, checking drift	0.20
Compliance and audit	Audits, legal review, regulator reporting	0.20
Total yearly cost	2.00 + 0.80 + 0.70 + 0.90 + 0.20 + 0.20 + 0.20	**5.00**

Table A12: Yearly benefits of the AI security platform (illustrative, ₹ crore)

Benefit item	How the figure is worked out (assumed)	Benefit (₹ crore)
Fraud losses avoided	Yearly fraud loss of ₹8.75 crore reduced by 40%: 8.75 × 0.40	3.50
Cheaper and shorter breaches	About 2 incidents caught earlier, saving ₹1 crore each: 2 × 1	2.00
Analyst time saved	12,000 hours saved by automation × ₹1,000 per hour = ₹1.2 crore	1.20
Penalties avoided	4% chance of a ₹20 crore penalty: 0.04 × 20 (expected value)	0.80
Customers kept (trust)	250 customers retained × ₹20,000 yearly value = ₹0.5 crore	0.50
Total yearly benefit	3.50 + 2.00 + 1.20 + 0.80 + 0.50	**8.00**

Source: . All amounts are assumed. IBM's global averages [2] were used only as a general guide to the kinds of savings possible.

Table A13: ROI calculation step by step

Measure	Calculation	Result
Net benefit	Total benefit − total cost = 8.00 − 5.00	₹3.00 crore
ROI (%)	(Benefits − Costs) ÷ Costs × 100 = (8 − 5) ÷ 5 × 100	60%
Benefit-cost ratio	Benefits ÷ Costs = 8 ÷ 5	1.6 (₹1.60 back for every ₹1 spent)
Payback period	Cost ÷ monthly benefit = 5 ÷ (8 ÷ 12) = 5 ÷ 0.667	About 7.5 months

Table A14: What if the benefits turn out lower or higher? (cost kept at ₹5 crore)

Case	Benefit (₹ crore)	Net benefit (₹ crore)	ROI (%)
Benefits only 50% of plan	8 × 0.5 = 4.0	4.0 − 5 = −1.0	−20%
Benefits only 70% of plan	8 × 0.7 = 5.6	5.6 − 5 = 0.6	12%

Case	Benefit (₹ crore)	Net benefit (₹ crore)	ROI (%)
As planned	8.0	8.0 − 5 = 3.0	60%
Benefits 120% of plan	8 × 1.2 = 9.6	9.6 − 5 = 4.6	92%